1.

$$VaR_{0.99} = \mu_L + \sigma_L N^{-1}(0.99)$$

$$VaR_{0.99} = 20(2.326)$$
$$\approx 46.53 \text{ million}$$

$$ES_\alpha = \mu_L + \sigma_L \frac{\phi(N^{-1}(\alpha))}{1-\alpha}$$

$$ES_{0.99} = 20\,\frac{\phi(2.326)}{0.01}$$

$$\approx 53.30 \text{ million}$$

2.

(a) prove $VaR_\alpha(L_1+L_2) \le VaR_\alpha(L_1) + VaR_\alpha(L_2)$

$P(L_1 \le 0) = 0.98$, then $VaR_{0.97}(L_1) = 0$ & $VaR_{0.98}(L_2) = 0$.

$P(L_1+L_2 \le 0) = 0.98^2 = 0.9604$
$P(L_1+L_2 \le 10) = 2(0.98)(0.02) = 0.0392$
$P(L_1+L_2 \le 20) = 0.02^2 = 0.0004$

For $\alpha \to 0.98$, Loss ≤ 10, CDF approach:

$$0.9604 + 0.0392 = 0.9996.$$

So, $VaR_{0.98}(L_1+L_2) = 10 > VaR_{0.98}(L_1) + VaR_{0.98}(L_2) = 0$
which is not subadditivity

(b) 0.99% VaR

$$VaR_{0.99}(L_1) = VaR_{0.99}(L_2) = 10$$

$$VaR_{0.99}(L_1+L_2) = 10$$

then $10 \le 10+10$

99% VaR is subadditivity

(c)

ES is subadditivity.

so, $ES_{0.98}(L_1) = ES_{0.98}(L_2) = 10$

Loss = 20, 0.0004

Loss = 10, 0.0196

so $ES_{0.98}(L_1 + L_2) = \frac{20(0.0004) + 10(0.0196)}{0.02} = 10.2$

thus

$10.2 \leq 10 + 10$

3.

(a)

$P(L_1 \leq l) = 0.99$

$0.99 - 0.9875 = 0.0025$

so must be in default loss distribution

Loss from 0 - 10, then

$0.9875 + 0.0125 \frac{l}{10} = 0.99$

$VaR_{0.99}(L_1) = \$2$ million

(b)

If both not default,

$L_{port} = -0.4$

and one of the loss is Loss = V, $V \sim Uniform(0, 10)$

$L_{port} = V - 0.2$

P of one default is

$$0.0125 + 0.0125 = 0.25$$

so 97.5%, Loss = -0.4 and

2.5%, Loss = 0.2 to 9.8

$$0.975 + 0.025\frac{l+0.2}{10} = 0.99$$

$$Var_{0.99}(L_{prot}) = \$5.8 \text{ million}$$

Compare $Var_{0.99}(L_1) + Var_{0.99}(L_2) = 2+2 = 4$

but $5.8 \geq 4$

Not subadditivity

(C)

$$ES_{0.99}(L_1) = \frac{2+10}{2}$$

$ES_{0.99}(L_1) = \$6$ million.

For bank port, 1% loss is 5.8 - 9.8 million

$$\text{so, } ES_{0.99}(L_{port}) = \frac{5.8+9.8}{2}$$

$ES_{0.99}(L_{port}) = \$7.8$ million

$$7.8 \leq 6+6$$

subadditivity

(4)

Not always correct, this statement

Because $L_2 \geq L_1$ in every scenario, two distribution may still yield the same quantile value at a certain quantile point

Ex: L_1

· 50% (0, 1)

50% $L_1 = 1$

$$L_2 = \begin{cases} 1, L_1 < 1 \\ 2, L_1 = 1 \end{cases}$$

For every scenario $L_2 \geq L_1$, ~~but~~ when $\alpha = 0.5$,

$$VaR_{0.5}(L_1) = 1$$
$$VaR_{0.5}(L_2) = 1$$

(5)

(a) Exponential Distribution of CDF is

$$F_L(l) = 1 - e^{-l/\theta}, \quad l \geq 0$$

ask:

$$F_L(VaR_\alpha) = \alpha$$

then

$$1 - e^{-VaR_\alpha/\theta} = \alpha$$

$$e^{-VaR_\alpha/\theta} = 1 - \alpha$$

take log

$$VaR_\alpha = -\theta \ln(1-\alpha)$$

Since Exp Distribution has memoryless property, greater than threshold mean is still θ

$$ES_\alpha = VaR_\alpha + \theta$$

sub:

$$ES_\alpha = \theta[1 - \ln(1-\alpha)]$$

(b) start $F_L(VaR_\alpha) = \alpha$

$$1 - VaR_\alpha^{-1/\theta} = \alpha$$

$$VaR_\alpha^{-1/\theta} = 1 - \alpha$$

so $VaR_\alpha = (1-\alpha)^{-\theta}$

For ES:

$$ES_\alpha = \begin{cases} \frac{VaR_\alpha}{1-\xi}, & 0 \le \xi < 1 \\ +\infty, & \xi \ge 1 \end{cases}$$

Why is ξ so important, because ξ control the thickness of the tail